

Xiang Ji, Ph.D.

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Assistant Professor, Department of Statistics

Iowa State University, Ames, IA, 50011

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Personal Website: <http://xiang-ji-ncsu.github.io/>

Google Scholar: <https://scholar.google.com/citations?user=YH4BbIMAAAJ&hl=en>



EDUCATION

Doctor of Philosophy in Bioinformatics and Statistics (Co-Major)	<i>December</i>
<i>North Carolina State University, Raleigh, NC</i>	<i>2017</i>
Dissertation: Phylogenetic Approaches for Quantifying Interlocus Gene Conversion	
Committee: Jeffrey Thorne, Eric Stone, Nadia Singh and Sujit Ghosh	
Master of Science in Material Science and Engineering	<i>September</i>
<i>North Carolina State University, Raleigh, NC</i>	<i>2013</i>
Thesis: Laser Interference Lithography for Fabrication of Gas Sensors	
Committee: John Muth, Lew Reynolds, Michael Kudenov and Jon-Paul Maria	
Bachelor of Science in Economics (Double Major)	<i>July 2011</i>
<i>Peking University, Beijing, China</i>	
Bachelor of Science in Physics	<i>July 2011</i>
<i>Peking University, Beijing, China</i>	
Advisor: Jia-Sen Zhang	

APPOINTMENTS

Assistant Professor	<i>Jan. 2026 – Present</i>
<i>Department of Statistics, Iowa State University, Ames, IA</i>	
Assistant Professor	<i>July 2020 – Dec. 2025</i>
<i>Department of Mathematics, Tulane University, New Orleans, LA</i>	
Faculty Investigator	<i>Nov. 2024 – Dec. 2025</i>
<i>Tulane Biostatistics and Bioinformatics Core</i>	
Faculty Member	<i>Oct. 2021 – Dec. 2025</i>
<i>Cancer Center, Tulane University, New Orleans, LA</i>	
Postdoctoral Fellow	<i>Jan. 2018 – July 2020</i>
<i>University of California, Los Angeles, CA</i>	

RESEARCH INTERESTS

Multigene family evolution; cancer systems biology; phylogenetics with large data sets; development of statistical techniques and software

PUBLICATIONS

58. Monti, F., **Ji, X.**, Shao, Y. and Suchard, M. A. (2026) Nonparametric Modeling of Continuous-Time Markov Chains. arXiv:2511.03954 [stat.ME][stat.CO] accepted by *Journal of the Royal Statistical Society Series B*
57. Swanepoel, C., Fourment, M., Ji, X., Nasif, H., Suchard, M. A., Matsen IV, F. A. and Drummond, A. J. (2026) TreeFlow: Probabilistic Modelling and Automatic Differentiation for Phylogenetics. *Systematic Biology*, syag072.
56. Wu, H., Yu, K., Xue, Y., Lu, M., Qi, Y., Deng, Z., Kong, S., Zhang, W., Ji, X., Baele, G., Li, B., Chen, Y., Jiang, Z., Wu, Z., Wang, W., Lu, Y. and Su, S. (2026) Spatial patterns of rodent-borne pathogens and ecological suitability of rodent-borne diseases in China. *Journal of Infection*, p.106824.
55. Shao, Y., Suchard, M. A., Rambaut, A., Ji, X., Lemey, P., Vasylyeva, T. I., & Baele, G. (2026). Parallel algorithms for phylogenetic inference under a structured coalescent approximation. *Proceedings of the National Academy of Sciences*, 123(18), e2602412123.
54. Gangavarapu, K., Ji, X., Shao, Y., Rambaut, A., Lemey, P., Baele, G. and Suchard, M. A. (2026) Tensor cores unlock efficient and lower-energy massive parallelization on phylogenetic trees. *Systematic Biology*
53. Fourment, M., Macaulay, M., Swanepoel, C.J., **Ji, X.**, Suchard, M.A. and Matsen IV, F.A. (2026) torchtree: flexible phylogenetic model development and inference using PyTorch. *Systematic Biology*, p.syaf047.
52. Zhang, M. N., Bao, H. and Ji, X. (2025) Development and validation of a novel palmitoylation-related prognostic signature in head and neck squamous cell carcinoma. *Translational Cancer Research*, 14(12), 8347-8364.
51. Baele, G., Carvalho, L. M., Brusselmans, M., Dudas, G., **Ji, X.**, McCrone, J. T., Lemey, P., Suchard, M. A. and Rambaut, A. (2025) HIPSTR: highest independent posterior subtree reconstruction in TreeAnnotator X. *Bioinformatics*.
50. Qi, K., Chen, J., Ma, X., Li, D., Baele, G., Li, X., Qin, Y., **Ji, X.**, Zhu, M., Li, X.X. and Guo, D. (2025) Novel Coronaviruses Identified in Livestock: The Urgent Need to Enhance Coronavirus Surveillance to Mitigate Zoonotic Risks. *Journal of Infection*.
49. Cong, F., Bao, H., Wang, X., Tang, Y., Bao, Y., Poulton, J., Liu, X., Wong, A., **Ji, X.** and Deng, W. (2025) Translocation of gut bacteria promotes tumor-associated mortality by inducing immune-activated renal damage. *The EMBO Journal*
48. Niu, Q., Jiang, Z., Wang, L., **Ji, X.**, Baele, G., Qin, Y., Lin, L., Lai, A., Chen, Y., Veit, M. and Su, S. (2025) Prevention and control of avian influenza virus: Recent advances in diagnostic technologies and surveillance strategies. *Nature Communications*
47. Baele, G., **Ji, X.**, Hassler, G. W., McCrone, J. T., Shao, Y., Holbrook, A. J., Lemey, P., Drummond, A., Rambaut, A., Suchard, M. A. (2025) BEAST X for Bayesian phylogenetic, phylodynamic and phylogeographic inference. *Nature Methods*
46. Pekar, J., et al., (2025) The recency and geographical origins of the bat viruses ancestral to SARS-CoV and SARS-CoV-2. *Cell*

45. Jiang, Z., Yan, Z., Hou, Y., Tang, J., Zheng, M., Lu, M., **Ji, X.**, Gangavarapu, G., Li, X., and Su, S., (2025). The RodentGPOMics Atlas: a comprehensive database of rodent biology for genomes and pathogens. *Nucleic Acids Research*.
44. Kang, M., Wang, L.F., Sun, B.W., Wan, W.B., **Ji, X.**, Baele, G., Bi, Y.H., Suchard, M.A., Lai, A., Zhang, M., ... Su, S., (2024). Zoonotic infections by avian influenza virus: changing global epidemiology, investigation, and control. *The Lancet Infectious Diseases*.
43. Magee, A. F., Holbrook, A. J., Pekar, J. E., Caviedes-Solis, I. W., Matsen IV, F. A., Baele, G., Wertheim, J. O., **Ji, X.**, Lemey, P., and Suchard, M. A., (2024). Random-effects substitution models for phylogenetics via scalable gradient approximations. *Systemic Biology*.
42. Yang, J., Schuemie, M.J., **Ji, X.** and Suchard, M.A., (2024). Massive parallelization of massive sample-size survival analysis. *Journal of Computational and Graphical Statistics*, 33(1), pp.289-302.
41. Gangavarapu, K., **Ji, X.**, Baele, G., Fourment, M., Lemey, P., Matsen IV, F.A. and Suchard, M.A., (2024). Many-core algorithms for high-dimensional gradients on phylogenetic trees. *Bioinformatics*, 40(2), p.btac030.
40. Huang, S., He, J., Yu, L., Guo, J., Jiang, S., Sun, Z., Cheng, L., Chen, X., **Ji, X.** and Zhang, Y., (2024). ASTK: A Machine Learning-Based Integrative Software for Alternative Splicing Analysis. *Advanced Intelligent Systems*, p.2300594.
39. Fisher, A., **Ji, X.**, Nishimura, A., & Suchard, M. A. (2023) Shrinkage-based random local clocks with scalable inference. *Molecular Biology and Evolution*, 40(11), msad242.
38. **Ji, X.**, Fisher, A. A., Su, S., Thorne, J. L., Potter, B., Lemey, P., Baele, G., & Suchard, M. A., (2023) Scalable Bayesian divergence time estimation with ratio transformations. *Systematic Biology*, syad039, arXiv preprint arXiv:2110.13298.
37. Yang, Y., Xu, T., Conant, G., Kishino, H., Thorne, J.L. and **Ji, X.**, (2023) Interlocus Gene Conversion, Natural Selection, and Paralog Homogenization. *Molecular Biology and Evolution*, msad198.
36. Matteson, N. L., et al., (2023) Genomic surveillance reveals dynamic shifts in the connectivity of COVID-19 epidemics. *Cell*, 186(26), pp.5690-5704.
35. He, W.T., Li, D., Baele, G., Zhao, J., Jiang, Z., **Ji, X.**, Veit, M., Suchard, M.A., Holmes, E.C., Lemey, P., Boni, M.F. and Su, S., (2023) Newly identified lineages of porcine hemagglutinating encephalomyelitis virus exhibit respiratory phenotype. *Virus Evolution*, 9(2), p.vead051.
34. Zhang, Z., Nishimura, A., Trovão, N.S., Cherry, J.L., Holbrook, A.J., **Ji, X.**, Lemey, P. and Suchard, M.A., (2023) Accelerating Bayesian inference of dependency between mixed-type biological traits. *PLOS Computational Biology*, 19(8), p.e1011419.
33. Zhao, J., Kang, M., Wu, H., Sun, B., Baele, G., He, W.T., Lu, M., Suchard, M.A., **Ji, X.**, He, N. and Su, S., (2023) Risk assessment of SARS-CoV-2 replicating and evolving in animals. *Trends in Microbiology*.
32. Zhao, J., Dellicour, S., Yan, Z., Veit, M., Gill, M.S., He, W.T., Zhai, X., **Ji, X.**, Suchard, M.A., Lemey, P. and Su, S., (2023) Early Genomic Surveillance and Phylogeographic Analysis of Getah Virus, a Reemerging Arbovirus, in Livestock in China. *Journal of Virology*, 97(1), pp.e01091-22.

31. Tsui, L.H., et al., (2023) Genomic assessment of invasion dynamics of SARS-CoV-2 Omicron BA. 1. *Science*, 381(6655), pp.336-343.
30. Hassler, G.W., Magee, A.F., Zhang, Z., Baele, G., Lemey, P., **Ji, X.**, Fourment, M. and Suchard, M.A., (2023) Data Integration in Bayesian Phylogenetics. *Annual Review of Statistics and Its Application*, 10, pp.353-377.
29. Fisher, A. A., Hassler, G. W., **Ji, X.**, Baele, G., Suchard, M. A., & Lemey, P. (2022) Scalable Bayesian phylogenetics. *Philosophical Transactions of the Royal Society B*, 377.1861(2022): 20210242.
28. Zhao, J., Dellicour, S., Yan, Z., Veit, M., Gill, M.S., He, WT, Zhai, X., **Ji, X.**, Suchard, M.A., Lemey, P. and Su, S., (2022) Early Genomic Surveillance and Phylogeographic Analysis of Gethah Virus, a Reemerging Arbovirus, in Livestock in China. *Journal of Virology*, pp.e01091-22.
27. Holbrook, A. J., **Ji, X.**, & Suchard, M. A. (2022) From viral evolution to spatial contagion: a biologically modulated Hawkes model. *Bioinformatics*, 38(7), pp.1846-1856. arXiv:2103.03348 [stat.ME]
26. McCrone, J.T., Hill, V., Bajaj, S. et al. (2022) Context-specific emergence and growth of the SARS-CoV-2 Delta variant. *Nature*, 610(7930), pp.154-160.
25. He, W., et al., (2022). Virome characterization of game animals in China reveals a spectrum of emerging pathogens. *Cell*, 185(7), 1117-1129.
24. Tian, J., Sun, J., Li, D., Wang, N., Wang, L., Zhang, C., Meng, X., **Ji, X.**, Suchard, M. A., Zhang, X., Lai, A., Su, S., & Veit, M. (2022) Emerging viruses: Cross-species transmission of Coronaviruses, Filoviruses, Henipaviruses and Rotaviruses from bats. *Cell Reports*
23. Zhao, J., Sun, J., Li, X., Xing, G., Zhang, Y., Lai, A., Baele, G. *, **Ji, X.** *, Su, S. * (2022). Divergent Viruses Discovered in Swine Alter the Understanding of Evolutionary History and Genetic Diversity of the Respirivirus Genus and Related Porcine Parainfluenza Viruses. *Microbiology Spectrum*, 10(3), pp.e00242-22. *Senior author
22. He, W., et al., (2022) Phylogeography reveals association between swine trade and the spread of porcine epidemic diarrhea virus in China and across the world. *Molecular Biology and Evolution*, 39(2), msab364
21. Fan, Y., et al. (2022). Systematic analysis of inflammation and pain pathways in a mouse model of gout. *Molecular Pain*, 18, 17448069221097760.
20. Xie, S., et al. (2022). Disrupted myelination network in the cingulate cortex of Parkinson's disease. *IET Systems Biology*, 16(3-4), pp.98-119.
19. Holbrook, A. J., **Ji, X.**, & Suchard, M. A. (2022) Bayesian mitigation of spatial coarsening for a fairly flexible spatiotemporal Hawkes model. *Annals of Applied Statistics*, 16 (1), 573-595.
18. Holbrook, A. J., Nishimura, A., **Ji, X.**, & Suchard, M. A. (2021). Computational Statistics and Data Science in the Twenty-first Century. *Wiley StatsRef: Statistics Reference Online*. arXiv preprint arXiv:2204.05530.
17. Lemey, P., et al., (2021) Untangling introductions and persistence in COVID-19 resurgence in Europe. *Nature*, 595 (7869), 713-717.

16. Landeros, A., **Ji, X.**, Lange, K., Stutz, T. C., Xu, J., Sehl, M. E., & Sinsheimer, J. S. (2021) An examination of school reopening strategies during the SARS-CoV-2 pandemic. *PloS One*, 16(5), e0251242.
15. Zhang, Z., Nishimura, A., Bastide, P., **Ji, X.**, Payne, R. P., Goulder, P., ... & Suchard, M. A. (2021). Large-scale inference of correlation among mixed-type biological traits with phylogenetic multivariate probit models. *The Annals of Applied Statistics*, 15(1), 230-251.
14. Lindelof, K., Lindo, J. A., Zhou, W., **Ji, X.**, & Xiang, Q. Y. J. (2020) Phylogenomics, biogeography, and evolution of the blue- or white-fruited dogwoods (Cornus) - insights into morphological and ecological niche divergence following intercontinental geographic isolation. *Journal of Systematics and Evolution*, 58(5), pp.604-645.
13. Sun, J., He, W. T., Wang, L., Lai, A., **Ji, X.**, Zhai, X., ... & Veit, M. (2020). COVID-19: epidemiology, evolution, and cross-disciplinary perspectives. *Trends in Molecular Medicine*, 26(5), pp.483-495.
12. He, W.*, **Ji, X.***, He, W. *, Dellicour, S. *, ..., & Su, S. (2020) Genomic epidemiology, evolutionary dynamics, and transmission patterns of porcine deltacoronavirus. *Molecular Biology and Evolution*, 37(9), pp.2641-2654. *equal contribution
11. **Ji, X.**, Zhang, Z., Holbrook, A., Nishimura, A., Baele, G., Rambaut, A., Lemey, P., & Suchard, M. A. (2020) Gradients *do* grow on trees: a linear-time O(N)-dimensional gradient for statistical phylogenetics. *Molecular Biology and Evolution*, 37(10), pp.3047-3060. arXiv:1905.12146 [stat.CO]
10. Fisher, A., **Ji, X.**, Zhang, Z., Lemey, P., & Suchard, M. A. (2020) Relaxed random walks at scale. *Systematic Biology*, 70(2), pp.258-267. arXiv:1906.04834[q-bio.PE]
9. Blestsa, M., Suchard, M. A., **Ji, X.**, Gryseels, S., Vrancken, B., Baele, G., Worobey, M., & Lemey, P. (2019) Divergence dating using mixed effects clock modelling: an application to HIV-1. *Virus Evolution*, 5(2), vez036.
8. Li, G., Zhang, W., Wang, R., Xing, G., Wang, S., **Ji, X.**, ... & Zhou, J. (2019). Genetic Analysis and Evolutionary Changes of the Torque teno sus Virus. *International journal of molecular sciences*, 20(12), 2881.
7. Zhou, W., **Ji, X.**, Obata, S., Pais, A., Dong, Y., Peet, R., & Xiang, Q., (2018) Resolving relationships and phylogeographic history of the Nyssa sylvatica complex using data from RAD-seq and species distribution modeling. *Molecular Phylogenetic and Evolution*, 126, 1-16.
6. **Ji, X.** (2017). Phylogenetic approaches for quantifying interlocus gene conversion. Doctoral Dissertation
5. **Ji, X.**, Griffing, A., & Thorne, J. L. (2016). A phylogenetic approach finds abundant interlocus gene conversion in yeast. *Molecular Biology and Evolution*, 33(9), 2469-2476.
4. Wang, K., Yu, S., **Ji, X.**, Lakner, C., Griffing, A., & Thorne, J. L. (2015). Roles of Solvent Accessibility and Gene Expression in Modeling Protein Sequence Evolution. *Evolutionary Bioinformatics*, 11, 85.
3. **Ji, X.** (2013). Laser Interference Lithography for Fabrication of Gas Sensors. Master Thesis
2. Han, X., **Ji, X.**, Wen, H., & Zhang, J. (2012). H-shaped resonant optical antennas with slot coupling. *Plasmonics*, 7(1), 7-11.

1. Xiao, G., **Ji, X.**, Gao, L., Wang, X., & Zhou, Z. (2012). Effect of dipole location on profile properties of symmetric surface plasmon polariton mode in Au/Al₂O₃/Au waveguide. *Frontiers of Optoelectronics*, 5(1), 63-67.

Articles in Submission and Preparation

9. Lemey, P., **Ji, X.**, Vrancken, B., Bletsa, M., Datta, P., Kafetzopoulou, L. E., Mifsud, J. C. O., Baele, G., Pourkarim, M. R., Patrono, L., Calvignac-Spencer, S., Orlando, L., Bastide, P., Guindon, S., Martin, D. and Suchard, M. A., (2026) Reconciling fast Hepatitis B evolutionary rates with ancient co-divergence. In revision with *PNAS*. Preprint on bioRxiv, doi:10.64898/2026.06.05.730483
8. Redelings, B., Wright, A., Gangavarapu, K., Shao, Y., Monti, F., Deng, W.-M., Su, S., Rambaut, A., Baele, G., Lemey, P., Suchard, M. and **Ji, X.**, (2026) BEAGLE in Action: Phylogenetic Likelihoods for Very Large State Spaces. In submission to *Systematic Biology*
7. Lemey, P., **Ji, X.**, et al., (2026) Reconciling fast Hepatitis B evolutionary rates with ancient co-divergence. bioRxiv, 2026.06. 05.730483
6. Pekar, J., et al., (2026) The emergence and molecular evolution of H5N1 influenza viruses in United States dairy cattle. In revision with *Cell*. bioRxiv, 2026.03.30.713641
5. Clessin, A., et al., (2026) Dispersal, adaptation and persistence of H5N1 in the sub-Antarctic and Antarctica. bioRxiv, 2026.03.20.713283
4. Monti, F., Faria, N. R., **Ji, X.**, Lemey, P., Kramer, M. U. G. and Suchard, M. A. (2026) Non-Linear Drivers of Population Dynamics: a Nonparametric Coalescent Approach. arXiv: 2602.06148 [stat.AP]
3. Bao, Y., Deng, W., Su, S., Lemey, P., Suchard, M. A., Glatt-Holtz, N.* and **Ji, X.*** (2025) Reflection Hamiltonian Monte Carlo for divergence time estimations. *Senior author. In preparation
2. **Ji, X.**, Redelings, B., Su, S., Bao, H., Deng, W., Monti, F., Hong, S. L., Baele, G., Lemey, P. and Suchard, M. A. (2025) Detecting Shifts in Natural Selection with Branch-Specific Substitution Models and Shrinkage Priors. In revision with *Systematic Biology*, arXiv:2507.08386 [q-bio.PE][stat.CO]
1. **Ji, X.**, Thorne, J. L. (2024) A phylogenetic approach disentangles the tract length and initiation rate of interlocus gene conversions. In wait/delay for submission, arXiv:1908.08608 [q-bio.PE]

TEACHING EXPERIENCE

Assistant Professor

Department of Statistics, Iowa State University, Ames, IA

- Fall 2026
 - **STAT 5101 Section 02 Statistical Methods for Data Analysis (4 credits)**
 ISU Canvas site: <https://canvas.iastate.edu/courses/131479>
 Level: Graduate
 Topics: randomization tests, two-sample inference (means and proportions), diagnostics, ANOVA and multiple comparisons, simple linear and logistic regression, multiple regression, factorial ANOVA and blocking

Enrollment: 39

- ***STAT 3030 Section 01 Probability and Statistics for Computer Science (3 credits)***

ISU Canvas site: <https://canvas.iastate.edu/courses/128800>

Level: Undergraduate

Topics: probability axioms, combinatorics, conditional probability, Bayes' rule, discrete distributions, joint distributions, continuous distributions, central limit theorem, Markov chains, Poisson process, estimation, confidence intervals, hypothesis testing, regression

Enrollment: 40

- Spring 2026

- ***STAT 4212/5212 Section 03 Experimental Design and Data Analysis (3 credits)***

ISU Canvas site: <https://canvas.iastate.edu/courses/126618>

Level: Undergraduate, Graduate

Topics: Sample size and power, Blocking, factorial treatment design, split plot design, repeated measures

Enrollment: 20

Assistant Professor

Department of Mathematics, Tulane University, New Orleans, LA

Summary: I regularly teach graduate-level Math 7260 (Linear Models) and Math 7360 (Data Analyses, a Tulane version of statistical learning), and undergraduate-level Math 3070 (Intro to Probability). Periodically, I teach Math 1230 Stats for Scientists, which is designed for students with non-math majors.

- Fall 2025

- ***Math 3070/6070 Intro to Probability (3 credits)***

GitHub site: <https://tulane-math-3070.github.io/2025Fall/>

Level: Undergraduate, Graduate (Master)

Topics: Axioms of probability, discrete and continuous random variables, multivariate distributions, expectation, limit theorem

Enrollment: 42

- Spring 2025

- ***Math 6040/7260 Linear Models (3 credits)***

GitHub site: <https://tulane-math-7260.github.io/2025Spring/>

Level: Graduate (Master + Ph.D.)

Topics: Simple and multiple linear regressions, hypothesis testing, analysis of variance, bootstrap, logistic regression

Enrollment: 7

- Fall 2024

- ***Math 1230 Stats for Scientists (3 credits)***

Level: Undergraduate

Topics: Probability, random variable, discrete and continuous distributions, sampling distributions, one and two-sample estimations, hypothesis testing, simple linear regression

- Enrollment: 62
- ***Math 3070/6070 Intro to Probability (3 credits)***
 GitHub site: <https://tulane-math-3070.github.io/2024Fall/>
 Level: Undergraduate, Graduate (Master)
 Topics: Axioms of probability, discrete and continuous random variables, multivariate distributions, expectation, limit theorem
 Enrollment: 33
- Spring 2024
 Teaching Relief
- Fall 2023
- ***Math 7360 Data Analysis (3 credits)***
 GitHub site: <https://tulane-math-7360-2023.github.io/>
 Level: Graduate (Master + Ph.D.)
 Topics: R, RMarkdown, Rcpp, git, data visualization using ggplot2, web scraping, linear models, generalized linear models, neural network, classification
 Enrollment: 22
- ***Math 3070/6070 Intro to Probability (3 credits)***
 GitHub site: <https://tulane-math-3070.github.io/2023Fall/>
 Level: Undergraduate, Graduate (Master)
 Topics: Axioms of probability, discrete and continuous random variables, multivariate distributions, expectation, limit theorem
 Enrollment: 48
- Spring 2023
- ***Math 6040/7260 Linear Models (3 credits)***
 GitHub site: <https://tulane-math-7260.github.io/2023Spring/>
 Level: Graduate (Master + Ph.D.)
 Topics: Simple and multiple linear regressions, hypothesis testing, analysis of variance, bootstrap, logistic regression
 Enrollment: 13
- ***Math 7980 Independent Study (3 credits)***
 Level: Graduate (Master)
 Topics: Item response theory models
 Enrollment: 1
- Fall 2022
- ***Math 3070/6070 Intro to Probability (3 credits)***
 GitHub site: <https://tulane-math-3070.github.io/2022Fall/>
 Level: Undergraduate, Graduate (Master)
 Topics: Axioms of probability, discrete and continuous random variables, multivariate distributions, expectation, limit theorem
 Enrollment: 37
- ***Math 1230 Stats for Scientists (3 credits)***
 Level: Undergraduate

Topics: Probability, random variable, discrete and continuous distributions, sampling distributions, one and two-sample estimations, hypothesis testing, simple linear regression

Enrollment: 66

- ***Math 7980 Independent Study (3 credits)***

Level: Graduate (Ph.D.)

Topics: Bayesian Phylogenetics

Enrollment: 1

- Spring 2022

- ***Math 6040/7260 Linear Models (3 credits)***

GitHub site: <https://tulane-math-7260.github.io/2022Spring/>

Level: Graduate (Master + Ph.D.)

Topics: Simple and multiple linear regressions, hypothesis testing, analysis of variance, bootstrap, logistic regression

Enrollment: 16

- ***Math 7980 Independent Study (3 credits)***

Level: Graduate (Master)

Topics: Review of optimization methods

Enrollment: 1

- Fall 2021

- ***Math 7360 Data Analysis (3 credits)***

GitHub sites: <https://tulane-math-7360-2021.github.io/>

Level: Graduate (Master + Ph.D.)

Topics: R, RMarkdown, Rcpp, git, data visualization using ggplot2, web scraping, linear models, generalized linear models, neural network, classification

Enrollment: 19

- Spring 2021

- ***Math 6040/7260 Linear Models (3 credits)***

GitHub site: <https://tulane-math-7260.github.io/2021Spring/>

Level: Graduate (Master + Ph.D.)

Topics: Simple and multiple linear regressions, hypothesis testing, analysis of variance, bootstrap, logistic regression

Enrollment: 17

- Fall 2020

- ***Math 7360 Data Analysis (3 credits)***

GitHub site: <https://tulane-math7360.github.io/>

Level: Graduate (Master + Ph.D.)

Topics: R, RMarkdown, Rcpp, git, data visualization using ggplot2, web scraping, linear models, generalized linear models, neural network, classification

Enrollment: 12

STUDENT ADVISING

Co-Supervisor (Doctoral)

*Department of Computational Biology, Institut Pasteur
Paris, France*

- Student name: John Tay
- Supervisor: Sebastian Duchene
- Research Title: TBD

2024 –
Present

Committee Chair (Master)

Department of Mathematics, Tulane University, New Orleans, LA

- Student name: Yufei Zou
- Research Title: Evolutionary inference with interlocus gene conversion

2023 –
Present

Committee Chair (Doctoral)

Department of Mathematics, Tulane University, New Orleans, LA

- Student name: Yuwei Bao
- Research Title: Reflective Hamiltonian Monte Carlo method

2022 –
2026

Dissertation Defense Committee Chair (Doctoral)

Department of Mathematics, Tulane University, New Orleans, LA

- Student name: Yuwei Bao
- Chair: Xiang Ji
- Research title: Reflective Hamiltonian Monte Carlo method

2026

Dissertation Defense Committee Member (Doctoral)

Department of Mathematics, Tulane University, New Orleans, LA

- Student name: John Argentino
- Chair: Michelle Lacey
- Research title: Multivariate Modeling Techniques for Estimating Assistance Impacts from Daily Food Security Surveys

2026

Prospectus Exam Committee Member (Doctoral)

Department of Mathematics, Tulane University, New Orleans, LA

- Student name: Sinchita Lahiri
- Chair: Samuel Punshon-Smith and Kyle Kun Zhao
- Research title: Analyzing the Decay Rates in PDE Systems

2025

Prospectus Exam Committee Member (Doctoral)

Department of Mathematics, Tulane University, New Orleans, LA

- Student name: John Argentino
- Chair: Michelle Lacey
- Research title: Multivariate Modeling Techniques for Estimating Assistance Impacts from Daily Food Security Surveys

2025

Dissertation Defense Committee Member (Doctoral)

Department of Mathematics, Tulane University, New Orleans, LA

- Student name: Linh Do
- Chair: Scott McKinley

2025

Research title: Statistical techniques for comparing populations of continuous piecewise-linear stochastic processes	
Prospectus Exam Committee Chair (Doctoral)	2024
<i>Department of Mathematics, Tulane University, New Orleans, LA</i>	
- Student name: Yuwei Bao - Chair: Xiang Ji - Research title: Scalable Bayesian statistical phylogenetics models	
Dissertation Defense Committee Member (Doctoral)	2024
<i>Department of Mathematics, Tulane University, New Orleans, LA</i>	
- Student name: Daniela A. Florez Pineda - Chair: Ricardo Cortez	
Oral Exam Committee Member (Doctoral)	2024
<i>Department of Mathematics, Tulane University, New Orleans, LA</i>	
- Student name: John V. Argentino - Chair: Michelle Lacey	
Prospectus Exam Committee Member (Doctoral)	2024
<i>Department of Mathematics, Tulane University, New Orleans, LA</i>	
- Student name: Daniela A. Florez Pineda - Chair: Ricardo Cortez - Research title: Mathematical models for transmission and control of mosquito-borne diseases	
Prospectus Exam Committee Member (Doctoral)	2023
<i>Department of Mathematics, Tulane University, New Orleans, LA</i>	
- Student name: Linh Do - Chair: Scott McKinley - Research title: Statistical techniques for comparing populations of continuous piecewise-linear stochastic processes	
Oral Exam Committee Member (Doctoral)	2023
<i>Department of Mathematics, Tulane University, New Orleans, LA</i>	
- Student name: Moslem Uddin - Chair: Lisa Fauci	
Oral Exam Committee Chair (Doctoral)	2023
<i>Department of Mathematics, Tulane University, New Orleans, LA</i>	
- Student name: Yuwei Bao - Chair: Xiang Ji	
Oral Exam Committee Member (Doctoral)	2023
<i>Department of Mathematics, Tulane University, New Orleans, LA</i>	
- Student name: Sinchita Lahiri - Chair: Kyle Kun Zhao	
Oral Exam Committee Member (Doctoral)	2022
<i>Department of Mathematics, Tulane University, New Orleans, LA</i>	

• Student name: Linh Do • Chair: Scott McKinley	
Oral Exam Committee Member (Doctoral)	2022
<i>Department of Mathematics, Tulane University, New Orleans, LA</i>	
• Student name: Daniela A. Florez Pineda • Chair: Ricardo Cortez	
Independent Study Advisor (Master)	2022
<i>Department of Mathematics, Tulane University, New Orleans, LA</i>	
• Student name: Paul Pluscht • Project topic: Review of optimization methods	
Undergraduate Research Advisor (Undergraduate)	2020 –
<i>Department of Statistics, North Carolina State University, Raleigh, NC</i>	2022
• Student name: Yixuan Yang • Co-advisor: Jeffrey L. Thorne • Research topic: Quantifying interlocus gene conversion and natural selection	
Undergraduate Research Advisor (Undergraduate)	2021
<i>Department of Mathematics, Tulane University, New Orleans, LA</i>	
• Student name: Jiayun Ling • Co-advisor: Xin Jiang • Research topic: Item response theory model, Add Health data, social science	

MENTEE ACHIEVEMENTS

Megan (Jiayun) Ling (undergraduate research advisor)	2021
• Oral presentation at ASA Louisiana Chapter meeting in 2021 • Graduated from master's program in Data Analytics of Duke University • Now working as Data Analyst at Blizzard Entertainment	
Yuwei Bao (PhD thesis advisor)	2022 –
• Oral presentation at ASA Louisiana Chapter meeting in 2022 • Oral presentation at Tulane Math Graduate Student Colloquium in 2022 • Oral presentation at Evolution Meeting 2023 • Oral presentation at Scientific Computing Around Louisiana Meeting 2023 • Oral presentation at Math for All in Nola meeting 2023 • Poster presentation at New England Statistical Symposium 2024 • Poster presentation at Pharmaceutical Data Science Conference online 2024 • Poster presentation at ASA Conference on Statistical Practice 2024 • Tuition Scholarship & Travel Award at Summer Institute in Statistical Genetics 2023 • Travel Award from Society for Study of Evolution at Evolution Meeting 2023 • Summer Research Fund from Math Department at Tulane University 2023 • Travel Award from Graduate Studies Student Association at Tulane University 2023 & 2024	2026

- Student Poster Award at New England Statistical Symposium 2024
- FDA-OCE-ASA Oncology Educational Fellow 2024
- Internship at Moderna Spring 2025
- Biostatistician at Foundation Medicine starting July 2026

Yufei Zou (undergraduate & PhD research advisor)

2022 –

- Tuition Scholarship & Travel Award at Summer Institute in Statistical Genetics 2023
- Oral presentation at ASA Louisiana Chapter Meeting on May 19, 2025
- Oral presentation at Evolution Meeting 2025
- SSE Caregiver Awards for attending Evolution Meeting 2025
- Attending PhD program in Data Science at UNC Charlotte
- Poster presentation at Data Science Across North Carolina 2025 at UNC Chapel Hill

AWARDS

HPC Fund Research Cloud Allocation Award ¹	AMD Corporation	2023
NVIDIA Academic Hardware Grant ²	NVIDIA Corporation	2022
Tuition Scholarship	SISMID ³	2018
NIEHS Fellowship ⁴	North Carolina State University	2015
SAMSI Graduate Fellow	SAMSI ⁵	2014
Tuition Scholarship	SISG ⁶	2013
University Graduate Fellowship	North Carolina State University	2011

FUNDED PROJECTS

External

National Institute of Health

04/01/26 –

- Title: Statistical innovation to integrate sequences and phenotypes for scalable phylodynamic inference
- Principle Investigator: Marc A. Suchard
- Subaward Investigators: Xiang Ji
- Total Amount: \$1,833,000
- Shared Amount: \$204,805

03/31/31

Louisiana Board of Regents Research Competitiveness Subprogram

07/01/23 –

- Title: Molecular epidemiology through scalable statistical phylogenetic Modelling
- Principle Investigator: Xiang Ji
- Total Amount: \$159,000

06/30/26

¹ 1,080 node hours (targeting 4 MI100 GPUs/node)

² Donation of one A100 80Gb PCIE GPU card

³ SISMID: Summer Institute in Statistics and Modeling in Infectious Diseases at the University of Washington at Seattle

⁴ The funds were matched through North Carolina State University

⁵ SAMSI: The Statistical and Applied Mathematical Sciences Institute

⁶ SISG: Summer Institute in Statistical Genetics at the University of Washington at Seattle

- Direct Cost: \$159,000

National Science Foundation

05/01/23 –
07/31/24

- Title: Evolutionary inference with interlocus gene conversion
- Principle Investigator: Jeffrey L. Thorne
- Co-Investigator: Xiang Ji
- Award number: DEB1754142
- Total Amount: \$564,338
- Shared Amount: \$94,351

National Institute of Health

07/01/22 –
03/31/25

- Title: Statistical innovation to integrate sequences and phenotypes for scalable phylodynamic inference
- Principle Investigator: Marc A. Suchard
- Subaward Investigator: Xiang Ji
- Award number: R01AI153044
- Total Amount: \$2,321,335
- Shared Amount: \$117,259

National Institute of Health

04/01/22 –
03/31/25

- Title: Notch signaling and germline-soma interactions in *Drosophila* ovarian model
- Principle Investigator: Wu-Min Deng
- Co-Investigator: Xiang Ji
- Award number: R01GM072562
- Total Amount: \$ 2,432,208
- Shared Amount: \$86,460

National Institute of Health

02/01/20 –
01/31/22

- Title: Consortium for viral systems biology (CViSB)
- Principle Investigator: Robert F. Garry
- Co-Investigator: Xiang Ji
- Award number: U19AI135995
- Total Amount: \$1,143,554
- Shared Amount: \$51,774

Internal

Tulane University CoR Research Fellowship

05/01/23 –
05/31/24

- Title: Scalable molecular epidemiology models
- Principle Investigator: Xiang Ji
- Total Amount: \$5,500

SUBMITTED PROPOSALS

National Science Foundation

2026

- Title: CAREER: Statistical Models and Parallel-Computing Methods for Analyzing Large-Scale Phylogenomic Datasets
- Principle Investigator: Xiang Ji
- Total Amount: \$583,798
- Status: Pending

National Science Foundation

2025

- Title: FEC: Identifying factors that mitigate the effect of poverty on child brain development using computational neural fingerprinting techniques
- Principle Investigator: Yu-ping Wang
- Co-Investigators: Xiang Ji
- Total Amount: \$4,000,000
- Shared Amount: \$20,000
- Status: Not Funded

National Institute of Health

2025

- Title: COBRE for Basic and Translational Research in Cancer (BATRIC)
- Principle Investigator: Wu-Min Deng
- Core Co-Director: Xiang Ji
- Total Amount: \$11,420,264
- Shared Amount: \$86,460
- Status: Not Funded

National Science Foundation

2025

- Title: FEC: Identifying factors that mitigate the effect of poverty on child brain development using computational neural fingerprinting techniques
- Principle Investigator: Yu-ping Wang
- Co-Investigators: Xiang Ji
- Total Amount: \$4,000,000
- Shared Amount: \$20,000
- Status: Not Funded

National Institute of Health

2025

- Title: Scalable statistical advances to quantify interlocus gene conversion in multigene family evolution
- Principle Investigator: Xiang Ji
- Co-Investigators: None
- Total Amount: \$1,833,000
- Status: Pending

National Science Foundation

2024

- Title: Scalable phylogenomic inference through massively parallelized action-based BEAGLE library
- Principle Investigator: Xiang Ji

- Co-Investigators: Robert F. Garry and Marc A. Suchard
- Total Amount: \$1,600,775
- Status: Not Funded

National Institute of Health

2024

- Title: Tracking and controlling viral outbreaks through scalable biologically realistic phylodynamic integrations
- Principle Investigator: Xiang Ji
- Co-Investigators: Robert F. Garry and Marc A. Suchard
- Total Amount: \$2,693,909
- Status: Not Funded

National Institute of Health

2024

- Title: Characterizing intra- and inter-host Nov evolution using human challenge study samples
- Principle Investigator: Pengbo Liu
- Subaward Investigators: Xiang Ji
- Shared Amount: \$120,018
- Status: Not Funded

CONFERENCE PRESENTATIONS

Oral Presentations

- 15 **Ji, X.** (2026, July) *Action-based phylogenetic likelihood calculations for large state-space models.* ISBA-EAC conference, Kunming, Yunnan, China
- 14 **Ji, X.** (2026, June) *Scalable divergence time estimation via Hamiltonian Monte Carlo sampling.* ICSA Applied Statistics Symposium, Arlington, VA
- 13 **Ji, X.** (2026, May) *Detecting evolutionary change-points with branch-specific substitution models and shrinkage priors.* ASA Midwest Regional Conference in Statistics and Data Science, University of Minnesota, Minneapolis, MN
- 12 **Ji, X.** (2025, October) *Scalable phylogenetic inference with Hamiltonian Monte Carlo method.* Statistics and AI in Precision Health and Biology, AMS Sectional Meeting, Tulane University, New Orleans, LA
- 11 **Ji, X.** (2025, August) *Quantitative traits on massive trees with many incomplete measurements.* JSM 2025, Nashville, TN
- 10 **Ji, X.** (2025, June) *Action-based phylogenetic likelihood calculations for large state-space models.* Evolution 2025 Meeting, University of Georgia, Athens, GA
- 9 **Ji, X.** (2025, June) *Scalable divergence time estimations with Hamiltonian Monte Carlo method.* 2025 International Indian Statistical Association Conference, University of Nebraska-Lincoln, Lincoln, NE
- 8 **Ji, X.** (2025, May) *Scalable evolutionary analysis with Hamiltonian Monte Carlo methods.* STATGEN 2025: Conference on Statistics in Genomics and Genetics, ASA Section on Statistics in Genomics and Genetics, University of Minnesota, Minneapolis, MN

- 7 **Ji, X.** (2025, May) *Action-based phylogenetic likelihood calculations for large state– space models*. 2025 Spring ASA Louisiana Chapter Meeting, LSU Health Science Center, New Orleans, LA
- 6 **Ji, X.** (2023, June) Scalable phylogenetic inference with Hamiltonian Monte Carlo method. Evolution 2023 meeting, Albuquerque, NM
- 5 **Ji, X.** (2023, March) *Scalable phylogenetic inference*. SCALA 2023: Scientific Computing Around Louisiana, Tulane University, New Orleans, LA
- 4 **Ji, X.** (2021, August) *Divergence time estimation with Hamiltonian Monte Carlo sampling and ratio transform*. Invited speaker at 2nd AsiaEvo Conference, Online
- 3 **Ji, X.** (2021, July) *Divergence time estimation with Hamiltonian Monte Carlo sampling and ratio transform*. Society of Molecular Biology and Evolution Meeting, Online
- 2 **Ji, X.** (2018, September) *Large-scale molecular epidemiology for viruses: efficient algorithms and new models*. Oral session presented at the Taishan Forum on Viral Infectious Diseases, Taishan Medical University, Tai'an, Shandong Province, China
- 1 **Ji, X.** (2017, August) *Phylogenetic approaches for quantifying interlocus gene conversion*. Quest for Ortholog 5 Meeting, University of Southern California, Los Angeles, CA

Poster Presentations

- 3 **Ji, X.** (2026, May) *Action-based phylogenetic likelihood calculations for large state-space models*. STATGEN 2026, Emory University, Atlanta, GA
- 2 **Ji, X.** (2017, July). *Phylogenetic approaches for quantifying interlocus gene conversion*. Society of Molecular Biology and Evolution Meeting, Austin, TX
- 1 **Ji, X.** (2014, June). *A Phylogenetic approach for quantifying interlocus gene conversion*. Evolution Meeting, Raleigh, NC

INVITED TALKS

16	Scalable phylogenetic Hamiltonian Monte Carlo method and its application in molecular epidemiology <i>Iowa State University, Ames, IA</i>	03/05/2025
15	Scalable phylogenetic Hamiltonian Monte Carlo method and its application in divergence time estimations <i>Southeastern Louisiana University, Hammond, LA</i>	08/29/2024
14	Phylogenetic approach for estimating amounts of interlocus gene conversion in duplications <i>AMS/AWM at Tulane University, New Orleans, LA</i>	12/06/2023
13	Scalable phylogenetic inference via Hamiltonian Monte Carlo method <i>Indiana University, Bloomington, IN</i>	02/03/2023
12	Scalable phylogenetic algorithm, modeling, and inference <i>North Carolina State University, Raleigh, NC</i>	10/10/2022
11	Smooth non-parametric coalescent priors for scalable divergence time estimations <i>AMS/AWM at Tulane University, New Orleans, LA</i>	10/05/2022
10	Divergence time estimation with Hamiltonian Monte Carlo sampling and ratio transform	08/05/2020

	<i>KU Leuven, Leuven, Belgium</i>	
9	Scalable modeling and inference for phylogenetics – from interlocus gene conversion to evolving pathogens <i>Tulane University, New Orleans, LA</i>	12/04/2019
8	Scalable modeling and inference for phylogenetics – from interlocus gene conversion to evolving pathogens <i>Louisiana State University, Baton Rouge, LA</i>	11/04/2019
7	Large-scale molecular epidemiology for viruses <i>University of California, Los Angeles, CA</i>	11/08/2018
6	Phylogenetic approaches for quantifying interlocus gene conversion <i>Zhejiang University, Hangzhou, China</i>	09/18/2018
5	Phylogenetic approaches for quantifying interlocus gene conversion <i>Chinese Academy of Sciences, Beijing, China</i>	09/04/2018
4	Phylogenetic approaches for quantifying interlocus gene conversion <i>CGM Online</i>	05/30/2018
3	Phylogenetic approaches for quantifying interlocus gene conversion <i>Duke University, Durham, NC</i>	02/15/2018
2	Phylogenetic approaches for quantifying interlocus gene conversion <i>Temple University, Philadelphia, PA</i>	04/25/2017
1	Phylogenetic approaches for quantifying interlocus gene conversion <i>North Carolina State University, Raleigh, NC</i>	03/24/2017

SOFTWARE

IGCexpansion

- My software for studying interlocus gene conversion.
- Available at <https://github.com/xji3/IGCexpansion>.

IGC BEAST Tutorial

- A tutorial for the Bayesian method (under development) of quantifying interlocus gene conversion in BEAST.
- Available at https://github.com/xji3/IGC_BEAST_Tutorial.

BEAST X

- I am a developer of the BEAST X software package.
- Available at <https://github.com/beast-dev/beast-mcmc>.

BEAGLE

- I am a developer of the BEAGLE library.
- Available at <https://github.com/beagle-dev/beagle-lib>.

Bito

- I collaborate with a group of statisticians and programmers to help them use my linear-time gradient algorithm implementations for their variational Bayesian phylogenetics development.
- Available at <https://github.com/phylovi/bito>.

PROFESSIONAL MEMBERSHIPS

American Statistical Association (lifetime), The International Society for Bayesian Analysis (lifetime), International Chinese Statistical Association (lifetime), Society for Systematic Biologists (lifetime)

SYNERGISTIC ACTIVITIES

Grant Panelist

- I served as a guest panel member for the UK MRC Better Methods, Better Research (BMBR) Panel in 2023.

Professional service

- I serve as an Editorial Board member for *Scientific Reports* by Springer Nature since March 2026.
- I serve as an Editorial Board member for *Discover Viruses* by Springer Nature since June 2024.
- I serve as a Review Editor for *Frontiers in Statistical Genetics and Methodology* since March 2021.
- I have reviewed manuscripts for
 - *Nature Medicine* (3)
 - *Proceedings of National Academy of Sciences* (2)
 - *Molecular Biology and Evolution* (9)
 - *Systematic Biology* (1)
 - *Annals of Applied Statistics* (2)
 - *Journal of Computational and Graphical Statistics* (1)
 - *Bioinformatics* (1)
 - *Theoretical Population Biology* (1)
 - *The American Journal of Human Genetics* (1)
 - *Plos Pathogens* (1)
 - *Briefings in Bioinformatics* (1)
 - *Microbiology Spectrum* (1)
 - *IEEE/ACM Transactions on Computational Biology and Bioinformatics* (1)
 - *Frontiers in Public Health* (2)
 - *Frontiers in Virology* (2)
 - *BMC Ecology and Evolution* (1)
 - *Evolutionary Bioinformatics* (2)
 - *Scientific Reports* (1)
 - *International Journal of Data Mining and Bioinformatics* (1)
 - *Rapid Reviews: COVID-19* (1)
 - *Open Veterinary Journal* (1)

Departmental service

- I am serving on the Seminar Committee of the Statistics Department at Iowa State University since Fall 2026.

- I am a faculty member of the consulting team in the Statistics Department at Iowa State University since Spring 2026.
- I led the development and establishment of a new Master of Science in Data Science program jointly hosted by the Math and Computer Science Departments at Tulane University.
- I served as the Director of Master Programs of the Math Department at Tulane University from Fall 2021 to Spring 2024.
- I served on the Graduate Study Committee of the Math Department at Tulane University from Fall 2021 to Spring 2024.
- I served on the Search Committee for a Professor of Practice in Statistics position from Fall 2021 to Spring 2023.
- I organize the Prob & Stats research seminar series of the Math Department at Tulane University since Fall 2023.
- I ran a tutorial workshop on building personal website for graduate students in the Math Department on 04/05/2022 (tutorial link: xji3.github.io/Tutorial).
- I serve on the School of Science & Engineering Cypress HPC committee as the representative for the Math Department in Spring and Summer 2025.

Outreach

- I served as a panelist for the “Preparing Data for Statistical Analysis” workshop in Grad Mixer Wednesday on 03/11/2026 at Iowa State University.
- I served as a poster judge for the CALS research day poster competition on 02/03/2026 at Iowa State University.
- I served as an ASN student poster award judge for the Evolution 2025 meeting.
- I participated in the LA FIRST Lego League State Championship as a volunteer judge in January 2023.
- I participated in Boys at Tulane in STEM (BATS) and Girls in STEM at Tulane (GiST), organized by Tulane Center for K-12 System Education
 - Volunteer, Boys at Tulane in STEM (BATS), 03/18/2023
 - Volunteer, Boys at Tulane in STEM (BATS), 09/16/2023
 - Volunteer, Girls in STEM at Tulane (GiST), 11/04/2023
 - Volunteer, Girls in STEM at Tulane (GiST), 03/09/2024
 - Volunteer, Boys at Tulane in STEM (BATS), 04/13/2024
 - Volunteer, Boys at Tulane in STEM (BATS), 09/21/2024
 - Volunteer, Girls in STEM at Tulane (GiST), 10/26/2024
 - Volunteer, Boys at Tulane in STEM (BATS), 02/01/2025
 - Volunteer, Girls in STEM at Tulane (GiST), 03/22/2025
 - Volunteer, Boys at Tulane in STEM (BATS), 09/20/2025
 - Volunteer, Girls in STEM at Tulane (GiST), 10/11/2025
- I served as treasurer of the ASSIST⁷ Student Leadership Council in 2012 and 2013. I participated in the Magnet Fair at South Raleigh Magnet High School as an ASSIST center graduate representative in 2012.

⁷ ASSIST: The NSF Center for Advanced Self-Powered Systems of Integrated Sensors and Technologies at NC State University